

The sources cover major concepts in Hypothesis and Inference, Distribution-Free Tests, Chi-Squared Tests, Correlation, and Simple Linear Regression.

Here is a detailed markdown summary of the important topics, points, and formulas:

I. Hypothesis and Inference

A. Core Concepts and Definitions

- A **Hypothesis** is an assumption about a population parameter.
- A **parameter** is a characteristic of the population, such as its mean or variance, and must be identified before analysis.
- **Hypothesis tests** are closely related to **confidence intervals**; generally, whenever one can be computed, the other can also be performed, and vice versa.
- A hypothesis test produces a number between 0 and 1 that measures the degree of certainty regarding the truth of a hypothesis about a quantity like a population mean or proportion.

Hypothesis Type	Representation	Description
Null Hypothesis	H_0	The claim that the effect indicated by the sample is due only to random variation between the sample and the population. In scientific research, it is the claim that the effect being studied does not exist.
Alternate Hypothesis	H_1	Says that the effect indicated by the sample is ** real** , accurately representing the whole population. It includes all possible alternatives other than the null hypothesis.

B. Types of Hypothesis Tests (Tailed Tests)

The type of test is determined by how the Null (H_0) and Alternate (H_1) hypotheses are stated:

Sl No.	Type of Hypothesis Test	Alternate Hypothesis (H_1)	Null Hypothesis (H_0)
1	Left Tailed Test	$<$	$\geq$
2	Right Tailed Test	$>$	$\leq$

Sl No.	Type of Hypothesis Test	Alternate Hypothesis (H_1)	Null Hypothesis (H_0)
3	Two Tailed Test	$\neq$	$=$

C. P-value Approach

- The **P-value** measures the plausibility of H_0 .
- The P-value is the probability of obtaining data as extreme as, or more extreme than, what was observed, **assuming H_0 is true**.
- The smaller the P-value, the stronger the evidence is against H_0 .
- **If the P-value is sufficiently small, we reject H_0 and believe H_1 instead.**
- The P-value is **not** the probability that H_0 is true, as H_0 either is true or is not true.

P-value Rules (Comparing with α):

- If $P \leq \alpha \rightarrow$ **Reject H_0** .
- If $P > \alpha \rightarrow$ **Fail to Reject H_0** .
- α is the **level of significance**. Some people use the "**5% rule**" ($P \leq 0.05$) to reject H_0 .

Calculating P-value in Large-Sample Tests for Mean (μ) or Proportion (p):

Alternate Hypothesis	P-value (Area under the normal curve)
$H_1 : \mu > \mu_0$ (or $H_1 : p > p_0$)	Area to the right of z
$H_1 : \mu < \mu_0$ (or $H_1 : p < p_0$)	Area to the left of z
$H_1 : \mu \neq \mu_0$ (or $H_1 : p \neq p_0$)	Sum of the areas in the tails cut off by z and $-z$

D. Test Selection Based on Sample Size (n) and Population Standard Deviation (S.D.)

Sample Condition	Population S.D. Known	Population S.D. Unknown
$n < 30$ (Small sample, Normal population)	Use z-test	Use t-test
$n \geq 30$ (Large sample)	Use z-test	Use z-test or t-test

E. Fixed-Level Testing and Errors

- **Fixed-level hypothesis testing** controls the probability of a Type I error (α).
- Setting α (e.g., 0.01, 0.05, or 0.10) provides a standard threshold for consistent decision-making.

Error Type	Definition (Condition)	Probability
Type I Error (False Positive)	Wrongly reject H_0 when it is true .	α (Level of significance).
Type II Error (False Negative)	Fail to reject H_0 when it is false .	β .
Correct Decision (H_0 True)	Accept H_0 when true.	$1 - \alpha$.
Correct Decision (H_0 False)	Reject H_0 when false.	$1 - \beta$ (Power).

F. Statistical Power

- The **Power of a test** is the probability of **correctly rejecting a false null hypothesis**.
- Power measures the test's ability to **detect a real effect** when one exists.
- A high-powered test reduces the risk of a Type II error (β).
- Tests with power greater than **0.80** or **0.90** are generally considered acceptable.
- Power analysis is performed before gathering data (to determine minimal sample size) and after gathering data (to determine magnitude of power).
- Statistical power has relevance only when the null hypothesis is false.

Computing Power (Two Steps):

1. Compute the **rejection region**.
2. Compute the probability that the test statistic falls in the rejection region if the alternate hypothesis (H_1) is true. This is the power.

Factors Affecting Statistical Power:

Factor	Relationship with Power	Detail
Sample size (n)	Larger n , Higher Power.	Increasing n is one way to increase power.
Significance level (α)	Larger α , Higher Power.	A larger α increases the probability of rejecting H_0 .
Effect size ($\mu_A - \mu_0$)	Greater Effect size, Greater Power.	If μ is far from H_0 , power is large; if μ is close to H_0 , power is small.

Factor	Relationship with Power	Detail
Standard deviation (σ)	Smaller σ , Greater Power.	Controlled by sampling from a homogeneous population or reducing measurement error.
Type of Statistical Test	One-tailed test Power $>$ Two-tailed test Power.	A one-tailed test effectively raises the significance level.

G. Formulas for Parametric Testing (Z-scores)

- **Z-score for Large Sample Mean Test (Implied):**

$$z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

Example calculation z (Large Sample Test for Population Mean):

$$z = \frac{1000.6 - 1000}{0.258} = 2.32$$

- **Z-score for Large Sample Proportion Test (Implied):**

$$z = \frac{x - np}{\sqrt{npq}}$$

Example calculation z (Coin Toss unbiased test):

$$z = 1.6$$

II. Distribution-Free (Nonparametric) Tests

A. General Points

- These methods are used when there is **no idea about the population distribution** from which the data is sampled.
- They are used for **small sample sizes**.
- They are also used when data are measured on an **ordinal scale** (ranks are meaningful), or when data are skewed, heavily tailed, or contain outliers.
- Distribution-free tests are sometimes called nonparametric tests.
- They do not involve population parameters.

B. Wilcoxon Signed-Rank Test

- A test for a **population mean**.
- The test uses the sum of the ranks corresponding to positive differences (S^+) and negative differences (S^-).
- To compute ranks, first subtract the hypothesized mean (μ_0) from each observation to obtain differences. The difference closest to 0 (ignoring sign) is assigned rank 1, and so on. Ranks corresponding to negative differences are given negative signs.

Wilcoxon Signed-Rank Formulas:

- The sum of positive ranks (S^+) and the absolute value of negative ranks (S^-) relationship:

$$S^+ + S^- = 1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

- **Large-Sample Approximation** ($n > 20$): S^+ is approximately normally distributed.
 - Mean of S^+ under H_0 :

$$\text{mean} = \frac{n(n+1)}{4}$$

- Variance of S^+ under H_0 :

$$\text{variance} = \frac{n(n+1)(2n+1)}{24}$$

- z -score:

$$z = \frac{S^+ - n(n+1)/4}{\sqrt{n(n+1)(2n+1)/24}}$$

- **Ties (zero differences):** Zero differences are **excluded** from ranking.

C. Wilcoxon Rank-Sum Test (Mann-Whitney U Test)

- A non-parametric test used to compare **two independent samples** to check if they come from the same population or if one tends to have larger values than the other.
- **Assumptions:** The populations must be continuous, and their probability density functions must be identical in shape and size, differing only in location.
- The test involves ordering the $m + n$ values from both samples combined and assigning ranks $1, 2, \dots, m + n$.

- The test statistic, denoted by $\mathbf{W}$, is the **sum of the ranks corresponding to the smaller sample** ($X_1, \dots, X_m$, where $m \leq n$).
- If $\mu_X < \mu_Y$, the rank sum W will tend to be smaller.

Wilcoxon Rank-Sum Large-Sample Approximation ($m > 8$ and $n > 8$):

- W is approximately normal.
 - Mean of W under H_0 :

$$\text{mean} = \frac{m(m + n + 1)}{2}$$

- Variance of W under H_0 :

$$\text{variance} = \frac{mn(m + n + 1)}{12}$$

- z -score:

$$z = \frac{W - m(m + n + 1)/2}{\sqrt{mn(m + n + 1)/12}}$$

III. Chi-Squared (χ^2) Test

A. Purpose

- Used to measure the **independence of two categorical variables** (i.e., if one variable affects the other).
- Also used to measure **goodness-of-fit** (i.e., if the observed and expected values match a prediction).
- Used when data is categorical.
- If the difference between observed and expected frequencies increases, the test statistic increases.

B. Chi-Squared Test Formulas

- **Expected Value Formula (for each cell):**

$$E = \frac{(\text{Row Total}) \times (\text{Column Total})}{\text{Grand total}}$$

- **Degrees of Freedom (for independence/homogeneity tests):**

$$df = (r - 1) \times (c - 1)$$

where r is the number of rows and c is the number of columns.

C. Types of Chi-Squared Tests

- **Test for Independence:** Used to test the null hypothesis that the probabilities of the column outcomes are the same for each row outcome.
- **Test for Homogeneity:** Used to test the null hypothesis that the distribution (proportions) of a categorical variable is the same across different populations (e.g., across different machines).

IV. Correlation and Simple Linear Regression

A. Data Classification

- **Univariate Data:** Analysis uses Central tendency measures (mean, median, mode), Dispersion (range, variance, standard deviation), and visualization (Histograms, Pie Charts).
- **Bivariate Data:** Analysis involves two variables (X and Y) represented as ordered pairs (X, Y) . X is the independent variable and Y is the dependent variable. Analysis uses Correlation Coefficient, Regression Analysis, and visualization (Scatter Plot).

B. Correlation and Pearson's Coefficient

- **Correlation** is the study of the relationship between two variables.
- **Scatter Plots** visually show the strength, direction, and presence of outliers in the relationship between variables.

Pearson's Correlation Coefficient (r) Formula:

- Measures the linear association between X and Y .
- It is the average of the product of z -scores.

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

Interpretation of r :

- $r = \pm 1 \implies$ Perfect Positive/Negative Correlation.
- $0 < r < 1 \implies$ Positive Correlation.

- $-1 < r < 0 \implies$ Negative Correlation.
- $r = 0 \implies$ No Correlation.

Properties and Cautions:

- The sample correlation (r) is used to measure strength and to construct confidence intervals and perform hypothesis testing on the population correlation (ρ).
- r is unitless.
- r remains unaltered if x and y are interchanged, a constant is added to a variable, or a variable is multiplied by a positive constant.
- The correlation coefficient measures **only Linear Association**. If the relationship is purely quadratic, r can be 0.
- r can be misleading when **outliers are present**.
- **Correlation is not causation**.

C. Simple Linear Regression (SLR)

- **Regression Analysis** is a set of statistical processes for estimating the relationships between a dependent variable (Y) and one or more independent variables (X).
- **Simple Regression Models** use one independent variable.
- The earliest form of regression analysis is linear regression, finding the line that most closely fits the data according to a mathematical criterion. This line is the **Least Squares Line**.

Least Squares Line Formulas:

- The equation of the Least Squares Line (the predicted value $\hat{y}_i$) is:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

Where $\hat{\beta}_0$ is the intercept and $\hat{\beta}_1$ is the slope.

- **Slope ($\hat{\beta}_1$):**

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

- **Y-Intercept ($\hat{\beta}_0$):**

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

- **Residual (e_i):** The vertical distance from a data point to the least-squares line.

$$e_i = y_{\text{observed}} - y_{\text{predicted}}$$

- The Least Squares Line is defined as the line for which the **sum of squared residuals** ($\sum e_i^2$) is **minimum**.

Goodness of Fit (r^2):

- A statistic that measures how well a model explains a given set of data.
- It is also referred to as the **coefficient of determination**.
- r^2 indicates how well a statistical model fits a data set (how close observed data are to the fitted line).
- It explains how much variation in the dependent variable Y is characterized by a variation in the independent variable X .
- Its value lies between **0** and **1**. The higher the value, the better the prediction.
- r^2 **Formula:**

$$r^2 = \frac{\sum_{i=1}^n (y_i - \bar{y})^2 - \sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

Note: $\sum_{i=1}^n (y_i - \bar{y})^2$ is the **Total sum of squares**.

D. Confounding Variables

- A confounding variable influences both the independent variable and the dependent variable, causing a **spurious correlation**.
- Confounding variables can increase variance or introduce bias.
- In **controlled experiments**, confounding can be avoided by choosing values for factors such that they are uncorrelated.
- In **observational studies**, confounding is difficult to avoid, and the study must be repeated multiple times under varied conditions before reliable conclusions are drawn.

V. Checking Assumptions and Transforming Data (SLR)

A. Assumptions for Errors (ϵ_i) in Linear Models

The errors $\epsilon_1, \dots, \epsilon_n$ must satisfy four assumptions for the least-squares estimates ($\hat{\beta}_0, \hat{\beta}_1$) to be useful:

1. The errors are **random and independent**.
2. The errors all have **mean 0**.
3. The errors all have the **same variance** (σ^2).
4. The errors are **normally distributed**.
 - *Note:* The normality assumption (4) becomes less important if the sample size is large.

B. Error Variance Estimate (σ^2)

- The vertical distance from each data point to the line is measured by the residual e_i .
- The estimate of the error variance σ^2 is s^2 , calculated using the sum of squared residuals.

C. Diagnostics using Residual Plots

- The single best diagnostic for least-squares regression is a plot of **residuals versus the fitted values** (a residual plot).
- If the linear model is valid and assumptions are met, the plot will show **no substantial pattern** (no curve, and the vertical spread should not vary much).

Condition	Description	Validity for Linear Model
Homoscedastic	Vertical spread in the scatterplot does not vary much.	Linear model is likely appropriate.
Heteroscedastic	Vertical spread varies (e.g., spread increases with fitted value).	Linear model is inappropriate .

D. Data Transformation

- If a residual plot shows a trend, curve, or is heteroscedastic, the assumptions for a linear model do not hold, and the data may need to be transformed.
- **Transformation** involves replacing a variable with a function of itself (e.g., power transformation, raising x or y to a power).
- Common transformations include **logarithm** ($\ln x$, $\ln y$) or using the **reciprocal** (power of -1).
- The goal is to find a transformation that results in a **homoscedastic residual plot with no discernible pattern**.

E. Outliers and Influential Points

- **Outliers** are points detached from the bulk of the data.

- Outliers should be checked to determine their cause (e.g., data-recording error, in which case they may be deleted).
- An **influential point** is an outlier that makes a considerable difference to the least-squares line when removed.
- Outliers with unusual x values are generally more likely to be influential than those with unusual y values.
- If transforming the data is unsuccessful, alternatives include:
 - **Weighted Least Squares:** Assigning greater influence to points where the error variance is smaller.
 - **Multiple Regression:** Adding more independent variables to explain the variation in the dependent variable.